

Forgetting Is a Changepoint: Anytime-Valid Retention Monitoring and E-Detector-Triggered Replay in Continual Learning

Abstract

Continual learning systems face a fundamental stability problem: learning new information can progressively destroy competencies acquired in the past. This phenomenon, commonly known as catastrophic forgetting, is typically evaluated retrospectively, after training, by measuring the degradation of performance on previously learned tasks. Such an approach is sufficient for benchmarking algorithms, but it is poorly suited to systems that must learn continuously and decide during training whether intervention is necessary.

This work studies catastrophic forgetting as an online statistical monitoring problem. Rather than asking only how much a model has forgotten at the end of training, we ask whether loss of retained competence can be detected sequentially, with explicit statistical guarantees, and whether such a detection mechanism can be used to control rehearsal.

We first formulate retained-task performance as a sequential risk process and investigate betting-based e-processes, which provide anytime-valid Type-I error control under continuous monitoring. Although statistically valid, the standard fixed-null e-process turns out to be structurally mismatched to forgetting. During periods in which retained-task risk remains comfortably below the tolerated threshold, the process loses betting wealth. Once forgetting begins, the accumulated deficit must first be recovered before an alarm can be raised. On a permuted-digits benchmark, this results in only 31.5% detection and a median detection latency of 257 SGD steps.

We therefore reinterpret forgetting as a changepoint problem and replace the fixed-start e-process with a restart-based e-detector. The detector effectively considers evidence accumulated from multiple possible changepoints and prevents historical evidence in favor of the null from indefinitely suppressing evidence of recent degradation. Detection rises to 94.5%, while median latency falls to 35 steps.

The stronger detection behavior comes with a different statistical guarantee: instead of controlling the probability of ever making a false rejection over an infinite horizon, the e-detector controls the expected run length to false alarm. We argue that this is the more appropriate guarantee when the monitor acts as a controller whose false alarms merely trigger unnecessary rehearsal.

Finally, we use the detector to schedule replay in a five-task continual-learning experiment. Triggered replay substantially outperforms naive fine-tuning and diagonal online EWC, but it does not outperform periodic replay when computational budgets are matched. Across several detector configurations, the triggered controller lies

approximately on the periodic replay accuracy–budget Pareto frontier. The principal benefit of the proposed approach on this benchmark is therefore not superior accuracy, but adaptive budget selection and statistically principled monitoring.

These results suggest a broader hypothesis: event-triggered rehearsal should provide an advantage primarily when interference is temporally concentrated. When forgetting evolves smoothly, uniform periodic rehearsal is already close to the allocation produced by the detector. This leads to a concrete and falsifiable research direction for larger continual-learning and language-model fine-tuning settings.

1. Introduction

Machine learning models are conventionally trained under an implicit assumption of temporal separation: training occurs first, deployment follows, and the learned parameters remain approximately fixed. Many real systems increasingly violate this assumption. Models may receive new tasks, new domains, new user data, new objectives, or additional fine-tuning long after their original training has finished.

This creates the central problem of continual learning. Improving performance on the present distribution can degrade performance on previous ones. In neural networks, gradient updates optimized for a new objective can overwrite parameter configurations that supported earlier capabilities. The resulting degradation is usually described as catastrophic forgetting.

A large literature attempts to reduce this effect through regularization, architectural isolation, memory buffers, gradient constraints, or replay. Yet most evaluations share a common structure: train the model, periodically evaluate old tasks, and report final accuracy or a forgetting metric.

There is a different question that matters when continual learning becomes an operational process:

Can we know, while training is occurring, when a previously learned capability has been damaged enough to require intervention?

This changes the problem from optimization alone to sequential statistical monitoring.

Suppose that after learning task T_k , a system retains a small monitoring set B_k . While the model learns subsequent tasks, examples from this set can be evaluated continuously. Each observation provides evidence about whether the original task remains sufficiently well learned.

The temptation is to repeatedly perform an ordinary hypothesis test. However, repeated testing creates an immediate statistical problem. A test calibrated to have a 5% false-positive probability at one predetermined time does not generally retain that property

when executed after every optimization step. With enough repeated looks at the data, random fluctuations eventually produce apparently significant evidence.

For a continual learner, this is not a minor technicality. Monitoring may occur hundreds, thousands, or millions of times.

Any useful monitoring procedure therefore needs to remain valid under optional stopping and continuous observation.

E-values and e-processes provide one possible solution. A non-negative e-process can be inspected at arbitrary times while retaining a uniform Type-I error guarantee through Ville's inequality. This makes e-processes attractive candidates for monitoring retained-task risk.

However, statistical validity alone does not imply that the instrument is well matched to the scientific question.

The central observation of this work is that catastrophic forgetting is fundamentally a changepoint phenomenon. Before interference occurs, a model may retain a task extremely well. At some later point, continued training changes the parameter configuration sufficiently that retained-task risk begins to exceed an acceptable level.

A monitor that accumulates evidence from the beginning of the process therefore carries potentially large amounts of historical evidence from the **pre-change regime**. This evidence becomes an obstacle after the change occurs.

The distinction leads to three main findings.

First, anytime-valid monitoring is empirically necessary. Repeated naive z-tests at nominal level $\alpha = 0.05$ produce a false-alarm probability of 45.3% over 3,000 observations, whereas the corresponding betting e-process produces 4.0%.

Second, the ordinary fixed-start e-process is poorly suited to detecting forgetting. A restart-based e-detector dramatically improves detection probability and latency.

Third, converting the detector into an event-triggered replay controller does not automatically yield superior continual-learning accuracy. On the benchmark studied here, it essentially recovers the same accuracy–compute frontier as periodic replay.

This negative result is informative. It suggests that the value of adaptive replay depends on the temporal structure of interference rather than merely on the existence of forgetting.

2. Continual Learning as Sequential Risk Monitoring

Consider a sequence of K learning tasks:

$T_1, T_2, \dots, T_k, \dots, T_K$

After training on task T_k , we retain a small labeled buffer B_k . In our experiments, each buffer contains 40 observations.

While training on subsequent tasks, one example is sampled from every retained buffer at each SGD step. For retained task k , define the binary random variable:

$$X_t^{(k)} \in \{0, 1\}$$

where $X_t^{(k)} = 1$ indicates that the current model makes an incorrect prediction on the sampled observation from B_k , while $X_t^{(k)} = 0$ indicates a correct prediction.

Because the parameters θ_t of the neural network evolve after every SGD update, the resulting sequence of observations cannot reasonably be assumed to be independent and identically distributed.

Instead, we define the instantaneous conditional retained-task risk as:

$$r_t^{(k)} = E[X_t^{(k)} | \mathcal{F}_{t-1}]$$

where $\mathcal{F}_{t-1}$ denotes all information available before observation t , including the previous training trajectory and monitoring observations.

The quantity $r_t^{(k)}$ therefore represents the expected instantaneous error of the current model on the retained-task buffer distribution.

After completing task T_k , we define an acceptable risk threshold:

$$r_0^{(k)} = \text{error on } B_k \text{ at the end of training } T_k + \delta$$

with:

$$\delta = 0.10$$

The sequential monitoring problem can then be expressed through the null hypothesis:

$$H_0^{(k)}: r_t^{(k)} \leq r_0^{(k)} \text{ for all } t$$

In words, task k is considered retained as long as its instantaneous conditional error does not exceed its post-training reference error by more than ten percentage points.

This formulation has an important consequence. Monitoring occurs while the underlying model itself changes. The data stream is therefore adaptively generated by the optimization process, making classical fixed-sample inference a poor fit.

2.1 Experimental environment

The experiments use the sklearn digits dataset, containing 1,797 observations represented by 64 input features. The dataset is divided using a 70/30 stratified train/test split.

Continual-learning tasks are generated by applying different fixed permutations to the input pixels. Detection experiments use $K = 2$ tasks, while the control experiments use $K = 5$ tasks.

The learner is a multilayer perceptron with architecture:

$64 \rightarrow 100 \rightarrow 10$

and ReLU activation.

Optimization uses stochastic gradient descent with:

learning rate = 0.05

batch size = 16

epochs per task = 6

Each task therefore corresponds to approximately 468 optimization steps.

The environment is intentionally small. The purpose of the benchmark is not to establish state-of-the-art continual-learning performance, but to isolate and understand the statistical behavior of the proposed monitoring mechanisms.

3. Why Ordinary Continuous Testing Fails

Suppose that retained-task performance is monitored using a conventional one-sided hypothesis test:

$H_0: r \leq r_0$

against:

$H_1: r > r_0$

with significance level:

$\alpha = 0.05$

If this test were conducted once at a predetermined sample size, its nominal Type-I error would be approximately 5%.

However, this interpretation breaks down when the same hypothesis is repeatedly tested after every SGD step.

Each new test represents another opportunity to observe an unusually large random fluctuation and incorrectly reject H_0 . Consequently, the probability of obtaining at least one false alarm can become much larger than α .

To quantify this effect, we consider a frozen model exactly at the null boundary:

$$r = r_0 = 0.10$$

The monitoring process is run for 3,000 sequential observations across 300 random seeds.

Repeated one-sided z-testing produces:

$$P(\text{false alarm over 3,000 steps}) = 0.453$$

Thus, a nominal 5% test incorrectly signals forgetting in approximately 45.3% of runs.

This is more than nine times the intended false-positive probability.

The result demonstrates that anytime validity is not merely a theoretical refinement. It becomes essential when a statistical test is embedded inside an iterative training process.

4. Anytime-Valid Monitoring with a Betting E-Process

A natural solution is to replace repeated fixed-sample tests with an anytime-valid e-process.

For binary observations $X_t \in \{0, 1\}$, consider the multiplicative process:

$$M_t = \prod_{i=1}^t [1 + \lambda_i(X_i - r_0)]$$

where the predictable betting fraction λ_i satisfies:

$$0 \leq \lambda_i \leq 1/r_0$$

Under H_0 , the conditional expectation of each multiplicative factor is:

$$E[1 + \lambda_i(X_i - r_0) \mid \mathcal{F}_{i-1}] = 1 + \lambda_i(r_i - r_0)$$

Since the null hypothesis implies:

$$r_i - r_0 \leq 0$$

we obtain:

$$E[1 + \lambda_i(X_i - r_0) \mid \mathcal{F}_{i-1}] \leq 1$$

Therefore, M_t is a non-negative supermartingale under H_0 .

Ville's inequality then provides the anytime-valid guarantee:

$$P(\sup_t M_t \geq 1/\alpha \mid H_0) \leq \alpha$$

For $\alpha = 0.05$, the rejection threshold becomes:

$$1/\alpha = 20$$

Therefore, forgetting is signaled whenever:

$$M_t \geq 20$$

regardless of how many times the process has previously been inspected.

Rather than choosing a single value of λ , the experiment uses a uniform mixture over the grid:

$$\lambda \in \{0.1, 0.2, \dots, 0.9\}/r_0$$

Since a mixture of non-negative supermartingales remains a supermartingale, this approach avoids selecting a single betting parameter while preserving anytime validity.

In the same 3,000-step null experiment described previously, the betting e-process produces:

$$P(\text{false alarm over 3,000 steps}) = 0.040$$

compared with:

$$P(\text{false alarm, naive z-test}) = 0.453$$

The e-process therefore remains below the nominal 5% level even under continuous monitoring.

This establishes the first empirical result of the study: anytime-valid inference is necessary for statistically reliable in-training monitoring.

5. Why the Fixed-Null E-Process Is the Wrong Object for Forgetting

Although the e-process solves the repeated-testing problem, a second and more subtle issue remains.

Imagine that immediately after completing task T_k , the model performs substantially better than the tolerated risk threshold:

$$r_t \ll r_0$$

During this period, the observed data provide evidence in favor of H_0 .

For a bettor attempting to detect the alternative $r_t > r_0$, the expected multiplicative growth is therefore below one. Consequently, the betting wealth M_t progressively decreases.

Now suppose that training on a new task eventually causes interference and the retained-task risk increases beyond r_0 .

The statistical regime has changed.

However, the e-process does not forget the evidence accumulated before this change.

Before it can cross the detection threshold, it must first recover the potentially enormous wealth deficit accumulated during the safe period.

This behavior is clearly visible in the experimental traces.

During the pre-forgetting phase, the base-10 logarithm of e-process wealth falls approximately to:

$$\log_{10}(M_t) \approx -8$$

At $\alpha = 0.05$, detection requires:

$$M_t \geq 20$$

or equivalently:

$$\log_{10}(M_t) \geq \log_{10}(20) \approx 1.30$$

The process may therefore need to recover roughly nine orders of magnitude of wealth before generating an alarm.

This explains why a statistically valid e-process can nevertheless be a poor detector of catastrophic forgetting.

The process itself is not broken. It is simply answering a different question.

The fixed-null e-process asks whether the accumulated observations since monitoring began provide sufficient evidence against the persistent null.

Forgetting instead asks whether the system has recently transitioned from an acceptable regime into an unacceptable one.

That is fundamentally a changepoint detection problem.

6. E-Detector with Restarts

To represent this changepoint structure, we replace the fixed-start e-process with a restart-based e-detector.

For each value of λ , the detector updates according to:

$$\log \tilde{M}_t = \max\{0, \log \tilde{M}_{t-1} + \log[1 + \lambda(X_t - r_0)]\}$$

The crucial operation is the maximum with zero.

Whenever accumulated evidence becomes favorable to the null, the detector is prevented from building an arbitrarily large negative balance. Operationally, this is analogous to restarting the betting process from a more recent candidate changepoint.

An equivalent interpretation is that $\tilde{M}_t$ tracks the strongest evidence obtainable from processes started at different points in the recent past.

The stopping time is:

$$\tau = \inf\{t : \tilde{M}_t \geq 1/\alpha\}$$

For $\alpha = 0.05$:

$$\tau = \inf\{t : \tilde{M}_t \geq 20\}$$

This apparently small modification radically changes detection behavior.

Across 200 seeds in the two-task experiment, the fixed-start e-process detects forgetting in only:

31.5% of runs

The restart-based e-detector detects forgetting in:

94.5% of runs

Median detection latency falls from:

257 SGD steps

to:

35 SGD steps

after the true test-error crossing.

The effect is especially pronounced for moderate forgetting.

For true forgetting between 0.10 and 0.20, detection probability increases from:

0.11 $\rightarrow$ 0.89

For forgetting between 0.20 and 0.30:

0.40 $\rightarrow$ 0.99

For forgetting of at least 0.30:

0.90 $\rightarrow$ 1.00

The experiment therefore demonstrates that the distinction between testing a persistent null and detecting a recent change is operationally substantial.

The fixed-start e-process is not statistically incorrect. It is structurally mismatched to the temporal phenomenon being monitored.

7. Two Different Guarantees for Two Different Purposes

The improved responsiveness of the e-detector comes at the cost of changing the statistical guarantee.

For the fixed-start e-process, Ville's inequality provides:

$$P(\exists t : \text{reject } H_0 \mid H_0) \leq \alpha$$

This statement holds over an arbitrarily long monitoring horizon.

In other words, if the null remains true forever, the probability that the process ever crosses its rejection threshold is bounded by α .

This is a particularly strong guarantee and is appropriate for one-shot certification.

The restart-based e-detector instead provides an average-run-length guarantee.

Let τ_{FA} denote the stopping time corresponding to a false alarm under the persistent null. Then:

$$E[\tau_{\text{FA}}] \geq 1/\alpha$$

For $\alpha = 0.05$:

$$E[\tau_{\text{FA}}] \geq 20$$

The two guarantees are not equivalent and should not be interpreted as such.

Which guarantee is preferable depends on the consequence of an alarm.

Suppose that the monitor is used to issue a formal statement:

“Subsequent training has not damaged task k .”

In this context, horizon-uniform Type-I error control is highly desirable because an incorrect rejection represents a false statistical conclusion.

Now consider a different use case in which an alarm simply causes the optimizer to execute several replay updates.

A false alarm then has a much smaller consequence: some additional computation is spent unnecessarily.

This distinction motivates separating two roles:

Certificate: use the fixed-start e-process.

Controller: use the restart-based e-detector.

Importantly, a practical system does not necessarily need to choose between them.

Both statistics can be maintained simultaneously using the same monitoring stream.

The e-process can provide conservative statistical certification, while the e-detector can control adaptive intervention.

8. From Detection to Triggered Replay

Once forgetting can be detected online, the next question is whether the detector can improve continual-learning behavior.

For each retained task k , we maintain an independent e-detector.

Whenever detector k fires, the system performs R SGD updates using the corresponding replay buffer B_k and subsequently resets the detector.

The experiments consider:

$R \in \{5, 20\}$

This creates an event-triggered rehearsal policy.

Instead of deciding in advance to replay every P optimization steps, the model spends replay computation only when the monitoring signal provides sufficient evidence of degradation.

The approach is compared with four baselines:

1. fine-tuning without replay;
2. diagonal online EWC;
3. periodic replay;
4. continuous replay as a high-compute reference.

Across five permuted tasks and 15 random seeds, ordinary fine-tuning obtains:

Average final accuracy = 0.570 ± 0.041

with forgetting:

0.469

The best diagonal online-EWC configuration, obtained with $\lambda = 10$, reaches:

Average final accuracy = 0.681 ± 0.033

Triggered replay with:

$R = 5$

$\alpha = 0.05$

$\delta = 0.10$

achieves:

Average final accuracy = 0.764 ± 0.019

using approximately:

207 replay steps

generated by approximately:

41 alarms

A periodic replay strategy matched to approximately the same computational budget uses 200 replay steps and reaches:

Average final accuracy = 0.758 ± 0.021

Continuous replay provides the highest performance:

Average final accuracy = 0.821 ± 0.012

with forgetting of:

0.143

However, this requires:

4,680 replay steps

which is more than twenty times the replay budget of the triggered controller.

At first sight, triggered replay therefore appears slightly better than periodic replay.

The paired difference is:

Triggered – Periodic = $+0.006 \pm 0.007$

for $R = 5$.

The same difference is observed for $R = 20$:

Triggered – Periodic = $+0.006 \pm 0.007$

The difference is not statistically significant.

Thus, although the detector successfully identifies forgetting and schedules meaningful interventions, the resulting adaptive replay does not provide convincing evidence of superior accuracy when compared with periodic replay at approximately the same computational budget.

9. Triggered Replay and the Accuracy–Budget Pareto Frontier

A stronger comparison requires considering the complete relationship between replay computation and final accuracy.

Periodic replay naturally defines an accuracy–budget trade-off.

Frequent rehearsal consumes more computation but generally preserves previous tasks more effectively. Infrequent rehearsal saves computation at the cost of greater forgetting.

The periodic replay sweep produces the following results:

$P = 25 \rightarrow \text{Budget} = 935 \rightarrow \text{Accuracy} = 0.813$

$P = 50 \rightarrow \text{Budget} = 460 \rightarrow \text{Accuracy} = 0.798$

$P = 100 \rightarrow \text{Budget} = 235 \rightarrow \text{Accuracy} = 0.772$

$P = 200 \rightarrow \text{Budget} = 110 \rightarrow \text{Accuracy} = 0.720$

$P = 400 \rightarrow \text{Budget} = 50 \rightarrow \text{Accuracy} = 0.649$

$P = 800 \rightarrow \text{Budget} = 20 \rightarrow \text{Accuracy} = 0.586$

These points define an empirical periodic accuracy–budget Pareto curve.

We then evaluate several triggered configurations.

For:

$(\alpha, \delta) = (0.05, 0.10)$

the controller self-selects a replay budget of:

207 steps

and reaches:

Accuracy = 0.764

The periodic frontier interpolated at approximately the same budget gives:

Accuracy ≈ 0.763

For:

$(\alpha, \delta) = (0.05, 0.05)$

triggered replay uses:

Budget = 245

and achieves:

Accuracy = 0.771

compared with approximately:

0.773

for periodic replay at the same budget.

For:

$(\alpha, \delta) = (0.20, 0.10)$

the triggered controller uses:

Budget = 307

and reaches:

Accuracy = 0.773

while the periodic interpolation is approximately:

0.78

Finally, for:

$(\alpha, \delta) = (0.01, 0.10)$

triggered replay uses:

Budget = 127

and achieves:

Accuracy = 0.736

compared with approximately:

0.73

for the corresponding periodic budget.

Across these four configurations, the detector automatically selects budgets spanning approximately a 2.4× range.

Yet in every case, the resulting point lies close to the periodic replay frontier.

The triggered controller therefore does not systematically shift the Pareto frontier upward.

This is arguably the most important negative result of the study.

The detector is useful, but not because it magically creates more accuracy from the same amount of replay.

Instead, its value lies in determining *when* replay is statistically justified and in selecting the effective replay budget adaptively.

10. Why Triggered Replay Does Not Beat Periodic Replay Here

Why does a statistically sophisticated event-triggered controller fail to outperform a simple periodic schedule?

The answer appears to lie in the temporal structure of forgetting.

Consider a fixed replay budget b .

Periodic replay distributes this budget approximately uniformly over the training trajectory.

If interference is also approximately uniform over time, then uniform rehearsal is already a sensible allocation strategy.

The retained-task traces in the permuted-digits experiment display approximately this behavior. Performance degradation tends to emerge progressively while the network learns subsequent tasks rather than occurring in isolated catastrophic bursts.

There is therefore relatively little temporal structure for the detector to exploit.

Triggered replay reallocates updates toward moments when evidence crosses the detector threshold, but the underlying need for rehearsal is not sharply concentrated around those moments.

Furthermore, sequential detection necessarily introduces some delay between the beginning of degradation and intervention.

This suggests a more general hypothesis.

Let:

g_t = instantaneous forgetting rate at time t

We can characterize the temporal burstiness of forgetting through the coefficient of variation:

$$CV(g) = SD(g_t) / E[g_t]$$

where $SD(g_t)$ is the standard deviation of the forgetting rate over time and $E[g_t]$ is its temporal mean.

When:

$$CV(g) \approx 0$$

the forgetting rate is approximately constant.

Interference is then smooth, and a uniform periodic allocation should already be close to optimal among simple replay strategies.

As $CV(g)$ increases, interference becomes increasingly concentrated into shorter temporal intervals.

In this regime, an event-triggered controller gains the opportunity to concentrate replay precisely where interference is strongest.

Define the accuracy advantage of triggered replay at budget b as:

$$\Delta(b) = A_{\text{triggered}}(b) - A_{\text{periodic}}(b)$$

The central empirical prediction is therefore:

$\Delta(b)$ should increase as $CV(g)$ increases

with:

$$\Delta(b) \approx 0 \text{ when } CV(g) \approx 0$$

The present permuted-digits benchmark appears to occupy the low-burstiness end of this spectrum.

This provides a possible mechanistic explanation for the null result and, more importantly, generates a falsifiable prediction for future experiments.

11. Implications for Continual Learning

The results suggest that catastrophic forgetting should be decomposed into two related but distinct problems.

The first is retention monitoring.

The system needs to determine whether a previously learned capability remains within an acceptable risk region.

The second is intervention scheduling.

Once evidence of degradation exists, the system must decide whether, when, and how strongly to rehearse previous information.

These problems need not share the same statistical machinery.

A horizon-uniform e-process is attractive when the objective is certification. A restart-based e-detector is more responsive when the objective is intervention.

This distinction becomes increasingly relevant as machine-learning systems move toward continuous post-deployment adaptation.

For example, consider a large language model that receives narrow domain-specific fine-tuning after deployment.

The system may need to preserve several pre-existing behavioral properties: factual competence, instruction following, safety behavior, calibration, or performance on strategically important domains.

Instead of evaluating these properties only after fine-tuning is complete, each could be associated with a bounded sequential monitoring signal.

The resulting architecture would contain multiple monitors operating simultaneously:

training stream → retained capability probes → sequential monitors → alarms → targeted intervention

Such an architecture transforms continual-learning protection from a fixed hyperparameter choice into a feedback-control problem.

Replay is no longer simply scheduled every P steps.

It becomes a response to measured evidence of capability degradation.

The current experiments show that this adaptivity does not guarantee better accuracy. Nevertheless, the framework provides a statistically interpretable connection between observed degradation and intervention.

12. Limitations

Several limitations prevent the current results from being interpreted as evidence for large-scale continual-learning systems.

First, the experiments operate at toy scale.

The dataset contains only 1,797 observations with 64-dimensional inputs, and the model is a shallow MLP. Phenomena that may dominate forgetting in large neural networks—representation superposition, feature reuse, safety-behavior drift, narrow fine-tuning damage, and complex cross-task transfer—are not represented.

Second, the control experiments use only 15 random seeds.

The observed standard errors of approximately 0.007 on the triggered-versus-periodic accuracy difference are sufficient to exclude large advantages, but not small ones.

Third, monitoring uses binary 0–1 error on a 40-sample buffer for each retained task.

Alternative bounded monitoring signals may provide substantially greater statistical efficiency. In particular, loss-based signals, calibration statistics, confidence-based quantities, or larger monitoring buffers deserve investigation.

Fourth, the continual-learning baselines are limited.

The EWC implementation uses an empirical diagonal Fisher approximation. Stronger approaches such as GEM, A-GEM, K-FAC-based approximations, or functional regularization were not included.

Finally, the benchmark itself appears to generate relatively smooth forgetting.

This is useful for understanding why triggered replay fails to outperform periodic replay, but it does not provide the regime in which the proposed controller is theoretically expected to be most useful.

13. Research Directions

The most important next experiment is therefore not simply to scale the same benchmark.

Instead, future experiments should manipulate the temporal structure of interference directly.

A benchmark family could be constructed in which the mean amount of forgetting is approximately fixed while its temporal concentration varies.

At one extreme, interference would occur continuously and smoothly.

At the other, the same total amount of degradation would occur through short, high-intensity bursts.

For each regime, one could measure:

$CV(g)$ = temporal coefficient of variation of forgetting

and:

$\Delta(b)$ = triggered accuracy – periodic accuracy at matched budget

The hypothesis is that the relationship between these quantities should be positive.

Class-incremental image-learning settings with sharp task boundaries provide one possible test environment.

Cyclic learning rates and warm restarts provide another mechanism for inducing temporally heterogeneous interference.

Large language model fine-tuning may provide an even more interesting regime.

A model could be subjected to short periods of narrow fine-tuning while a retained behavioral suite is monitored continuously. If degradation occurs abruptly around particular optimization phases, event-triggered intervention may allocate rehearsal more efficiently than uniform periodic replay.

A second research direction concerns the monitored quantity itself.

Accuracy is only one aspect of retained competence.

A model may maintain classification accuracy while becoming progressively less calibrated. In high-stakes settings, degradation in uncertainty estimation may matter before top-line performance visibly collapses.

Sequential monitoring could therefore be extended to calibration statistics and other bounded behavioral quantities.

A third direction concerns attribution.

The present framework assigns one detector to each retained task. More complex systems could maintain a factored bundle of detectors corresponding to tasks, domains, behavioral properties, or components of an agent.

An alarm could then provide not only evidence that the system has degraded, but information about where the degradation is occurring.

14. Conclusion

Catastrophic forgetting is conventionally framed as an optimization problem:

How can a model learn new information without damaging old information?

This work studies a complementary question:

How can we know, online and with statistical guarantees, that forgetting has begun?

The distinction matters.

Repeated fixed-sample hypothesis tests are unsuitable for continuous training-time monitoring. At a nominal significance level of $\alpha = 0.05$, naive sequential z-testing produces false alarms in approximately 45.3% of 3,000-step runs.

An anytime-valid betting e-process reduces the corresponding false-alarm probability to 4.0%.

Yet statistical validity alone is insufficient.

The fixed-start e-process is structurally mismatched to catastrophic forgetting because evidence accumulated during long periods of successful retention creates a wealth deficit that must be recovered after degradation begins.

In the two-task detection experiment, this results in:

Detection probability = 31.5%

and:

Median detection latency = 257 SGD steps

Treating forgetting explicitly as a changepoint produces a substantially different result.

The restart-based e-detector reaches:

Detection probability = 94.5%

with:

Median detection latency = 35 SGD steps

The price of this increased responsiveness is a change in statistical guarantee.

The fixed e-process provides horizon-uniform Type-I error control:

$$P(\exists t : \text{reject } H_0 \mid H_0) \leq \alpha$$

whereas the e-detector provides an average-run-length guarantee:

$$E[\tau_{\text{FA}}] \geq 1/\alpha$$

These guarantees correspond naturally to two different operational roles.

The e-process is appropriate for certification.

The e-detector is appropriate for control.

However, better detection does not automatically imply better continual-learning performance.

When the e-detector is used to trigger replay, the resulting controller achieves approximately the same final accuracy as periodic replay at matched computational budgets.

Across multiple values of α and δ , triggered replay lies approximately on the periodic accuracy–budget Pareto frontier rather than above it.

Its demonstrated advantage is therefore not superior accuracy.

Its advantage is that the replay budget is self-selected from observed degradation rather than specified through an externally tuned replay period, while the decision to intervene remains connected to a formal sequential statistical framework.

The negative result also suggests a more precise hypothesis about when adaptive replay should matter.

When forgetting is temporally smooth, periodic rehearsal distributes computation in approximately the same way that the underlying interference requires.

When forgetting is bursty, however, uniform rehearsal may waste computation during stable periods and underallocate it during periods of severe interference.

This leads to the central prediction:

the advantage of triggered replay over periodic replay at matched budget should increase with the temporal burstiness of forgetting.

The current benchmark appears to represent the low-burstiness regime, where this advantage is approximately zero.

The next step is therefore not merely to build a larger detector.

It is to test the mechanism.

If the predicted relationship between burstiness and adaptive-replay advantage holds across controlled continual-learning regimes—and eventually in narrow fine-tuning of large language models—then sequential changepoint detection could become more than a monitoring instrument.

It could provide the statistical layer of a closed-loop continual-learning system in which retained capabilities are continuously measured, degradation is detected online, and intervention is allocated precisely when the evidence indicates that it is needed.

The broader conclusion is simple:

Forgetting is not merely a final accuracy difference. It is a change in a stochastic process.

Once viewed in this way, continual learning becomes not only a problem of optimization, but also one of sequential inference, changepoint detection, and feedback control.

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